

Task-1

Decimal	GrayCode	Binary.
0	000	000
1	001	001
2	011	010
3	010	011
4	110	100
5	111	101
6	101	110
7	100	111

$G_0 G_1 G_2$

$B_0 B_1 B_2$

★ GrayCode to Bin : $G_0 = B_0$; $B_1 = B_0 \oplus G_1$; $B_2 = B_1 \oplus G_2$

★ $\begin{array}{c} a \\ +1 \text{ } \underline{2} \text{ } | b \\ e1 \text{ } \underline{1} \text{ } | c \\ d \end{array}$ $a, b, c, d, e, f, g \rightarrow$ o/p pins
where active = high (1).

★ PTO

→ Total 1 complete truth table.

GrayCode	decimal	a	b	c	d	e	f	g
000	0	1	1	1	1	1	1	0
001	1	0	1	1	0	0	0	1
011	2	1	1	0	1	1	0	1
010	3	1	1	1	1	0	0	1
110	4	0	1	1	0	0	1	1
111	5	1	0	1	1	0	1	1
101	6	1	0	1	1	1	1	1
100	7	1	1	1	0	0	0	1
[ABC]								

K map for each segment

segment a:

A \ BC	00	01	11	10
0	1	0	1	1
1	1	1	1	0

Groupings: $\overline{B}\overline{C}$ (top-left), $A\overline{B}$ (middle-left), BC (bottom-right), $\overline{A}B$ (top-right)

$$a = \overline{A}B + A\overline{B} + BC + \overline{B}\overline{C}$$

$$a = A \wedge B + \sim(B \wedge C)$$

segment b:

A \ BC	00	01	11	10
0	1	1	1	1
1	1	0	0	1

Groupings: $\overline{A}$ (top row), $\overline{B}$ (bottom-left), $\overline{C}$ (bottom-right)

$$b = \overline{A} + \overline{C}$$

$$b = \overline{AC} =$$

segment c:

A \ BC	00	01	11	10
0	1	1	0	1
1	1	1	1	1

Groupings: $\overline{C}$ (top-left), $\overline{B}$ (top-right), A (bottom-right), $\overline{C}$ (bottom-left)

$$c = A + \overline{B} + \overline{C}$$

$$c = \overline{BC} + A$$

segment d :

A \ BC	00	01	11	10
$\bar{A}\bar{B}\bar{C}$	0	1	1	0
$\bar{A}B\bar{C}$	0	1	1	0
$A\bar{B}\bar{C}$	0	0	0	0
$A\bar{B}C$	0	0	0	0
$A\bar{B}C$	0	0	0	0
ABC	0	0	0	0

$$\begin{aligned}
 d &= \bar{A}B + AC + \bar{A}\bar{B}\bar{C} \\
 &= \bar{A}(B + \bar{B}\bar{C}) + AC = \bar{A}(B + \bar{C}) + AC \\
 &= \bar{A}B + \bar{A}\bar{C} + AC = \bar{A}B + \sim(A \wedge C)
 \end{aligned}$$

segment e :

A \ BC	00	01	11	10
$\bar{A}\bar{B}\bar{C}$	0	1	1	0
$\bar{A}B\bar{C}$	0	1	1	0
$A\bar{B}\bar{C}$	0	0	0	0
$A\bar{B}C$	0	0	0	0
$A\bar{B}C$	0	0	0	0
ABC	0	0	0	0

$$\begin{aligned}
 e &= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C \\
 &= \bar{A}(\bar{B}\bar{C} + B\bar{C}) + A\bar{B}C \\
 &= \bar{A}\bar{C} + A\bar{B}C
 \end{aligned}$$

segment f :

A \ BC	00	01	11	10
$\bar{A}\bar{B}\bar{C}$	0	1	1	0
$\bar{A}B\bar{C}$	0	1	1	0
$A\bar{B}\bar{C}$	0	0	0	0
$A\bar{B}C$	0	0	0	0
$A\bar{B}C$	0	0	0	0
ABC	0	0	0	0

$$f = AC + AB + \bar{A}\bar{B}\bar{C} = A(B + C) + \bar{A}\bar{B}\bar{C}$$

segment g: A/BC

0
1

00

0

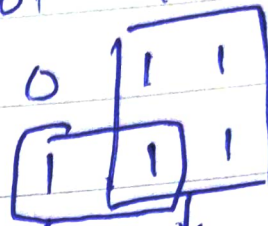
01

0

0

11

10



AC

B

$$g = AC + B.$$